PORTFOLIO ITAI 2376 · DEEP LEARNING SPRING 2026

Jiri Musil.

Coursework, labs, and projects completed in *Spring 2026*. Neural networks, transformers, diffusion, reinforcement learning, and a final multi agent system.

→ GITHUB REPOSITORY

github.com/JiriCZTX/Jiri_Musil_ITAI_2376_Deep_Learning

Public · README · Code · Notebooks · Slides

PROGRAM	Applied AI & Robotics, *HCC*
FOCUS	Multi-agent systems · NLP · forecasting
STACK	PyTorch · HF Transformers · CrewAI
STATUS	Open to roles & collaboration

• THE COURSEWORK

Eleven modules across the *deep learning* curriculum, with two capstone projects.

Spring 2026 · 16 weeks
Houston Community College
Instructor: Patricia McManus

L02	L03	A01	A02	L05	A03	A04	L08	L09	MID	FIN
Neural Networks from Scratch	Convolutional Networks	DL Tools & Frameworks	Neural Network Zoo	RNNs & Transformers	DL for an 11-Year-Old	"Arrival" Through NLP	Diffusion Models	Cart-Pole RL Agent	AI Agent Blueprint	Workforce Intelligence
LAB	LAB	ASSIGNMENT	ASSIGNMENT	LAB	GROUP · EXPLAINER	GROUP · ANALYSIS	LAB	LAB	MIDTERM	FINAL PROJECT

○ Lab notebook ○ Assignment ● Capstone

6+ HANDS-ON LABS

4 WRITTEN DELIVERABLES

2 CAPSTONE PROJECTS

Workforce Intelligence.
A *two agent* system for talent and attrition.

AGENT 01 · TALENT INTELLIGENCE

Routed NER ensemble + SBERT semantic match.

DistilBERT · GLiNER · ModernBERT

10 entity types · curated gazetteer

PROPRIETARY BRAIN

⇄

6 scenarios CrewAI + Claude (joint hire)

AGENT 02 · WORKFORCE FORECASTING

Bi-LSTM attrition & headcount projection.

Bi-LSTM attrition model

Temporal patterns · risk alerts

200 people
SYNTHETIC TEST WORKFORCE

8 depts
ENERGY INDUSTRY SCENARIO

840
ADJUDICATED DEV SPANS

3.1%
AVG MONTHLY ATTRITION TRACKED

A dashboard that *routes* entity recognition across DistilBERT, GLiNER, and ModernBERT, matches candidates to roles with SBERT, and forecasts department attrition with a Bi-LSTM.

→ 01 **Routed NER ensemble** · model selection per entity class

→ 02 **SBERT job to candidate matching** · cosine similarity over embeddings

→ 03 **Bi-LSTM attrition forecasting** · department headcount projections

→ 04 **Live dashboard** · industry switcher, adaptive learning engine

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Skills and tools *built* across the semester.

01
Deep *models*

CNN Image classification (LO3)

RNN / LSTM
Sequence learning (LO5)

Transformer
DistilBERT, ModernBERT, GLiNER

Bi-LSTM Attrition forecasting

Diffusion
DDPM image generation (LO8)

DQN Cart-Pole control (LO9)

02
NLP & *agents*

Named-entity recognition
10 entity types, routed ensemble

Sentence embeddings
SBERT for semantic match

Multi-agent orchestration
CrewAI + Claude

Prompting & tooling
6-scenario proprietary brain

Reinforcement learning
Policy & value methods

03
Frameworks *& tools*

PyTorch
Primary training framework

HF Transformers
Pretrained backbones

Sentence-Transformers
SBERT inference

Gymnasium RL environments

Pandas / NumPy
Data preparation

Streamlit / Plotly
Interactive dashboard

04
Practice *& craft*

Evaluation design
Adjudicated dev sidecar (840 spans)

Technical writing
Blueprints, READMEs, decks

Group collaboration
A03 & A04 group deliverables

Translating to non-experts
"DL for an 11-year-old"

Reproducible notebooks
Versioned, parameterised

• REFLECTION & NEXT STEPS

A semester of writing networks by hand, training them, and putting two agents to work.

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↗ WHAT'S NEXT

Goals for the next semester and beyond.

→ Extend Workforce Intelligence to public datasets

→ Continue reinforcement learning past Cart-Pole

→ ITAI 2373 (NLP) and ITAI 3377 (Edge AI and IoT)

→ Open to engineering, research, and collaborator roles

GITHUB

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CONTACT

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HCC · Applied AI & Robotics
Spring 2026